

03 Hours

[80 Marks]

N.B. a. Question no.1 is **compulsory**.

b. Attempt any three questions out of remaining five questions

c. Figures to right indicates full marks

- Q1 Attempt any **four** of the following [20]
- a. Describe automatic machines.
 - b. What is thread rolling?
 - c. How are seamless tubes manufactured?
 - d. Explain explosive welding
 - e. Differentiate between arc welding and electron beam welding.
- Q2. a. With a neat sketch explain the principle and working of laser beam welding. Also discuss its advantages, limitations and application. [08]
- b. Differentiate between the open loop and close loop system. [06]
 - c. Explain gas cutting operation. [06]
- Q3. a. Compare thermoplastics and thermosetting plastics. [08]
- b. Describe the desirable properties of moulding sand. [06]
 - c. What are transfer machines? [06]
- Q4. a. Describe types of drilling machines and there application. [08]
- b. Explain blow moulding process with a neat sketch. [06]
 - c. What is riser? Discuss its types and application. [06]
- Q5. a. Discuss various defects found in welding. [08]
- b. Define weldability and describe the factors affecting it. [06]
 - c. Differentiate between hot working and cold working processes. [06]
- Q6. a. What are the constituents of moulding sand? Discuss their function. [08]
- b. Describe machining the operations performed on lathe machine. [06]
 - c. Define the terms 'Spread', 'Elongation' and 'Draft' with respect to rolling process. [06]
